



Confidence in a connected world.

VERITAS Storage Foundation Cluster File System 5.1 from Symantec Clustered NFS.

*Karthik Ramamurthy, Technical Product Manager
David Noy, Sr. Product Manager
Cluster File System*

VERITAS Storage Foundation Cluster File System 5.1 from Symantec

Using CFS for NFS serving

Content

Introduction	3
About the Veritas Storage Foundation product line from Symantec.....	4
The NAS paradigm	5
CFS for scalable file serving	5
Clustered NFS (CNFS) architecture	6
CNFS protocol	7
DNS Round Robin	8
NFS advisory locking	8
TCP or UDP for transport.....	8
Configuring VCS resources and service groups	8
Configuring the CVM service group	10
Configuring the Clustered-NFS service group.....	11
Summary	13
Appendix	14

Introduction

The fundamental properties of CFS include access to shared file systems by applications running on all nodes of the cluster. One such application is network file serving (NFS).

The paper discusses the various aspects of NFS and how running NFS on CFS improves NFS scalability.

About the Veritas Storage Foundation product line from Symantec

Veritas Storage Foundation provides easy-to-use online storage management, enables high availability of data, optimized I/O performance, and allows freedom of choice in storage hardware investments. Veritas Storage Foundation is the base storage management offering from Symantec. It includes Veritas File System and Veritas Volume Manager. Both Veritas File System and Volume Manager include advanced features such as journaled file system, storage checkpoints, Dynamic Multi-Pathing, off-host processing, volume snapshots and tiered storage. Storage Foundation comes in three editions: Basic, Standard and Enterprise. Each targets different environments as described below:

Storage Foundation Basic is the freeware version of Storage Foundation. Available as a free download, it is limited to a maximum of 2 CPU and 4 volumes and 4 file systems. For more information, please visit: <http://www.symantec.com/business/theme.jsp?themeid=sfbasic>

Storage Foundation Standard is intended for SAN connected servers with high performance requirements and availability features, such as multiple paths to storage. This product is a minimum requirement for High Availability solutions.

Storage Foundation Enterprise includes the entire feature set of both File System and Volume Manager. It is designed for servers with large SAN connectivity, where high performance, off-host processing and tiered storage are desired.

http://www.symantec.com/business/products/overview.jsp?pcid=2245&pvid=203_1

Veritas Storage Foundation Cluster File System

Veritas Storage Foundation Cluster File System provides an integrated solution for shared file environments. The solution includes Veritas Cluster File System, Cluster Volume Manager and Cluster Server to help implement robust, manageable, and scalable shared file solutions. Veritas Cluster File System provides linear scalability for parallel applications and is widely used as a fast failover mechanism to ensure that application downtime is minimized in the event of server or software failure. With Veritas Storage Foundation Cluster File System, cluster-wide volume and file system configuration allows for simplified management; and extending clusters is simplified as new servers adopt cluster-wide configurations.

http://www.symantec.com/business/products/overview.jsp?pcid=2247&pvid=209_1

Veritas Cluster Server

Veritas Cluster Server is the industry's leading cross-platform clustering solution for minimizing application downtime. Through central management tools, automated failover, features to test disaster recovery plans without disruption, and advanced failover management based on server capacity, Cluster Server allows IT managers to maximize resources by moving beyond reactive recovery to proactive management of application availability in heterogeneous environments.

http://www.symantec.com/business/products/overview.jsp?pcid=2247&pvid=20_1

The NAS paradigm

File system access is becoming more broadly distributed between direct attached storage, file systems on SAN connected storage and multi-client access of files over the network, with networked file systems becoming increasingly popular.

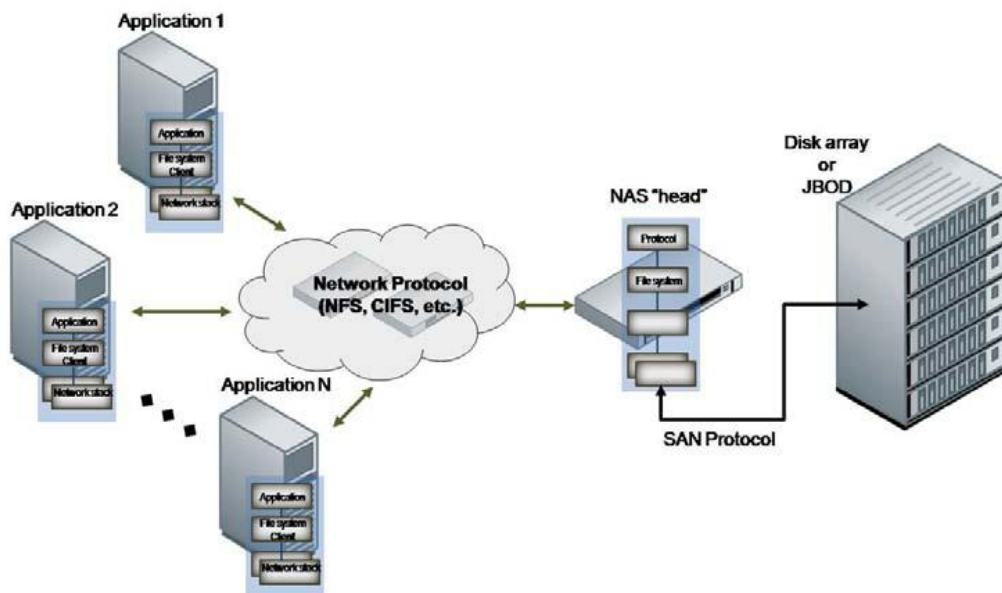


Figure 1: NAS model

The most frequently used multi-client file system technology is network-attached storage (NAS). The NAS head, which in smaller configurations is often integrated with disk drives, uses a file access network protocol, typically Network File System (NFS) or Common Internet File System (CIFS), to interact with client computers. However most NAS models have a couple of inherent limitations:

- Because of the higher level of protocol processing required, NAS systems are designed for higher access latency than storage access at the block level.
- The NAS head can be a potential bottleneck for I/O. Depending on the nature of the workload, one or more resources tends to become the bottleneck factor.

The bigger problem of NAS models is inflexibility. Typical NAS systems offer limited configurable options. They are limited in terms of scalability and an increase in either NFS client traffic or storage utilization can only be handled by procuring another NAS head or filer. Modular and independent scaling of client and storage capacity is expensive to achieve with NAS solutions.

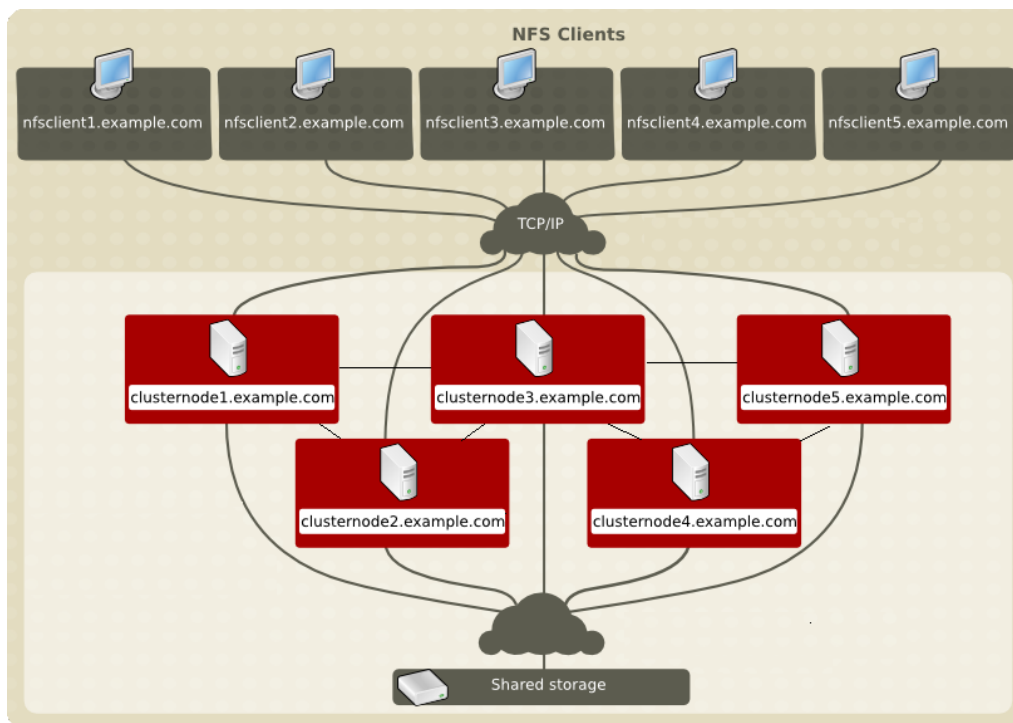
CFS for scalable file serving

The CFS architecture makes it an ideal platform for solving the limitations of a traditional NAS setup. It runs on all commercial UNIX flavors and Linux, from the very economical x64 processors to the multi-core enterprise class servers.

CFS is highly scalable and

- Supports up to 32 nodes in the cluster.
- It is built on top of Cluster Volume Manager which supports a wide range of storage arrays. For more details, refer to the HCL at <http://seer.entsupport.symantec.com/docs/283161.htm>
- The storage configuration is extensible, including multiple tiers with different cost and performance characteristics.
- The storage can be consumed either by FC or iSCSI.
- It supports high speed network connectivity between the nodes and to the clients, including high performance Ethernet interfaces.
- Any increase in the number of NFS clients is handled by adding a new node into the CFS cluster to serve the increased client load
- Any increase in storage requirements is handled by adding additional storage to the existing SAN infrastructure.
- It is a cluster variant of VxFS and benefits from all the capabilities of the file system.
- The VxFS File System enables effective exploitation of multi-tier storage through its Dynamic Storage Tiering (DST) facility. DST has two parts: support for multi-volume file systems and automatic policy-based placement of files within the storage managed by a file system. Multi-volume file systems, as the name implies, are file systems that occupy two or more virtual storage volumes. A VxFS multi-volume file system presents a single name space, making the existence of multiple volumes transparent to users and applications. But VxFS remains aware of each volume's identity, making it possible to control the locations at which individual files are stored.

Clustered NFS (CNFS) architecture



Clustered NFS is a solution to deliver active-active NFS serving over an underlying cluster file system. Each node in the CNFS clusters runs the complete CVM-CFS-VCS stack, and in addition, the CNFS server parallel application component. The CNFS server converts the NFS request from the client to POSIX file system requests, which it issues to the underlying CFS instance. The CFS and CVM instances cooperate as usual to provide coordinate concurrent access to one or more file systems from all the cluster nodes.

The cluster file system can be mounted on all the nodes in different modes, from read-only to read-write depending on the desired NFS exposure to clients. CNFS server implementation can be accomplished either as

- **Active-passive** – The primary server serves all NFS requests, the secondary monitors it and assumes the role of master if the primary fails.
- **Active-active** – This consists of two heads connected to common storage and both heads serve the same file system to clients. Should a head fail the other head recovers all the state associated with the NFS clients and resumes operations.

Note: *The first release of CNFS supports only active-active configurations, subsequent ones will support both active-active and active-passive.*

CNFS protocol

CNFS supports Version 3 of the NFS protocol (commonly called NFSv3) layered on the standard Remote Procedure Call (RPC) protocol. NFSv3 incorporates the eXternal Data Representation (XDR) standard for platform-independent data representation, making NFS client and server implementations completely interoperable regardless of platform type. NFSv3 RPCs can be transported by either UDP or TCP, although the latter is preferred. NFSv3 is *stateless*; servers do not retain information about clients between successive NFS operations. Statelessness simplifies recovery after a server failure, because all NFS operations are completely independent of any prior operations. The CNFS implementation of NFS includes two additional protocols required by many applications:

- The Mount protocol enables a server to restrict access to shared file systems, and integrates served file systems into client directory hierarchies
- The Lock protocol, implemented by the Network Lock Manager (NLM) supports advisory locking of entire files and ranges of bytes within files by clients.

Consolidating users' home directories on an NFS server centralizes and simplifies administration. A client computer can automatically mount an NFS server-hosted file system containing user home directories at a specific mount point so that each user's files are visible to him. Files in other users' home directories are accessible or not, depending on how the administrator configures home directories when creating them. If home directories move to another file system on the same or a different NFS server, remapping the automatic mount leaves users' views unchanged—scripts and other references to home directories continue to work unaltered.

DNS Round Robin

Before NFS clients connect to the CNFS server, the server's virtual host name and IP address must be registered in the DNS server. The clients connect to any of the CNFS servers using the virtual IP addresses. By using DNS to direct clients to the CNFS cluster, the clients are load balanced between all the nodes of the CNFS farm. The load balancing is taken care of by DNS and typically employs a round-robin mechanism to do so.

Before the CNFS cluster is put into operation, a network administrator must register the cluster's fully qualified domain name and list of virtual IP addresses with the enterprise DNS server(s), specifying round-robin response ordering. Since clients usually connect to the first IP address in a DNS list, round-robin ordering causes successive client connections to be distributed evenly among a cluster's nodes.

NFS advisory locking

The Network Lock Manager (NLM) is used to synchronize access to NFS files that are shared by other applications in the network. CNFS includes a cluster wide NLM service. Every node of the CFS cluster is capable of giving out locks. The lock states are maintained on a per client basis on a shared file system. In the event of a CFS server leaving the cluster, the locks are recovered on one of the other nodes. During this period of recovery, any further NFS lock requests either block or are dropped based on the request mode. By using the global lock manager which is part of CFS, CNFS ensures that no locks are granted during recovery from CFS server crash.

TCP or UDP for transport

Many if not all clients using NFS as a data store require reliable connectivity between the NFS client and the NFS server. For this reason, it is preferred to run NFS on top of TCP instead of UDP. Other than client reliability some of the other reasons to using TCP instead of UDP include

- The performance penalty with TCP is lower than that with UDP in case of recovery.
- TCP provides flow control and throttles back when a network becomes congested. UDP does not control traffic flow and can congest a network

For most UNIX and Linux environments, UDP is the default for NFS, so TCP must be explicitly specified in the mount commands.

Configuring VCS resources and service groups

The example in this document uses a two node CFS cluster with Clustered NFS configured in it with a single virtual IP.

The example configuration in this document uses the following setup

- **cvm** – Standard service group that controls the Veritas Cluster Volume Manager and the Cluster File System shared resources. This group is automatically created during the configuration phase of Storage Foundation Cluster File System installation. This service group manages CVM and the basic CFS functionality (a.k.a vxfsckd)
- **cfsnfssg** – This service group contains the CFS mount resources for the NFS share as well as the shared CFS mount resource needed for lock management. It consists of the NFS resource and the share resource apart from the CVMVoldg and CFSSMount resource.
- **vip1** – This service group contains the virtual IP and NIC resource needed for NFS clients to connect to. The virtual IP service group will fail over from one node to another during system failover. Typically more than one virtual IP is assigned per clustered-NFS cluster.

The **cvm** and **cfsnfssg** are configured as parallel service groups and are online on all the nodes. The **vip1** service group is configured as a failover service group.

The figure below illustrates how the different service groups within this solution depend on each other. For an in-depth description of the different types of service group dependencies, please see the Veritas Cluster Server Administrator's Guide.

(<http://seer.support.veritas.com/docs/286758.htm>)

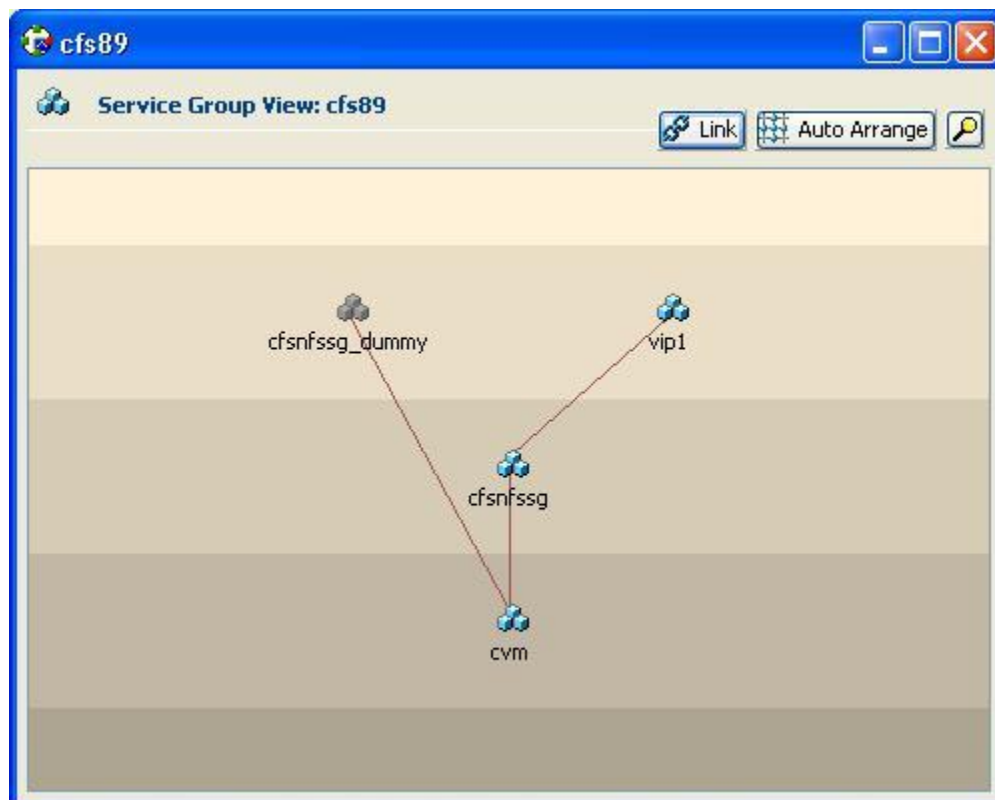


Figure 2: Showing the service groups

At the bottom of the dependencies tree the *cvm*-group provides the infrastructure for the Veritas Cluster Volume Manager and Cluster File System. This group will be the first to start, and the last to stop. The *cvm* group is a parallel group and will be online on all servers within the cluster during normal operations.

The *cfsnfssg* service group ensures availability of the clustered file systems and it has an “Online Local Firm” dependency on the *cvm* service group. This is to ensure that all required clustering components are functioning correctly before proceeding.

The *cfsnfssg* group also contains the resources needed to make NFS a parallel resource and mount points for lock recovery. The share resource to export file systems to NFS clients is part of this service group.

The *vip1* service group contains the failover virtual IP which the NFS clients connect to.

The *cfsnfssg_dummy* service group is a placeholder service group for NFS shares that are unshared or for CFS mounts that need to be deleted later.

Configuring the CVM service group

In all Storage Foundation Cluster File System clusters there should be a service group named *cvm*. This group is configured during installation and provides control and monitoring facilities for the Cluster Volume Manager and Cluster File System infrastructure resources. These resources must run on each system participating in a cluster. The VCS resources *cvm_vxconfigd*, *cvm_clus* and *vxfsckd* are mandatory services.

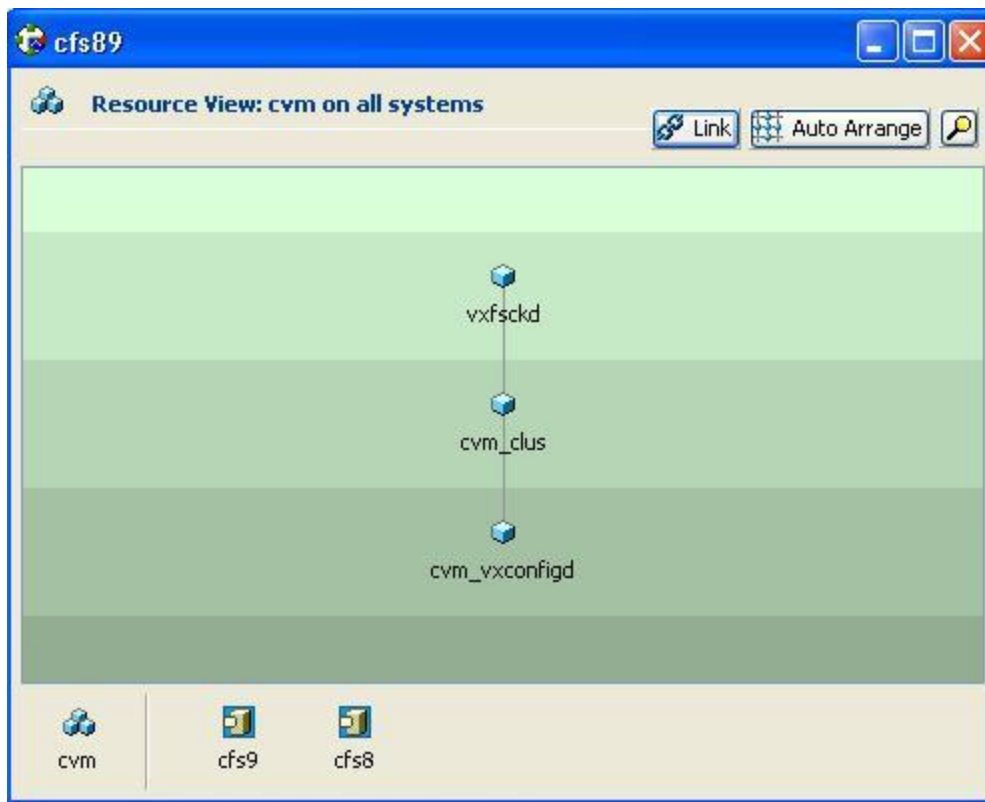


Figure 3: Default CVM Service Group

Configuring the Clustered-NFS service group

The clustered NFS service group contains the CVMVoldg resource used to online and monitor the shared volumes. The CFS mount is created on top of the volume and is monitored using the CFSSMount agent. The NFS agent monitors the NFS daemon. A share is created on top of the CFS mount for external clients to use. The Share resource depends on both the CFSSMount and the NFS daemon to be online.

A separate CFS mount is created for storing all the NLM locks shared in the cluster. This mount is also online on all the nodes simultaneously.

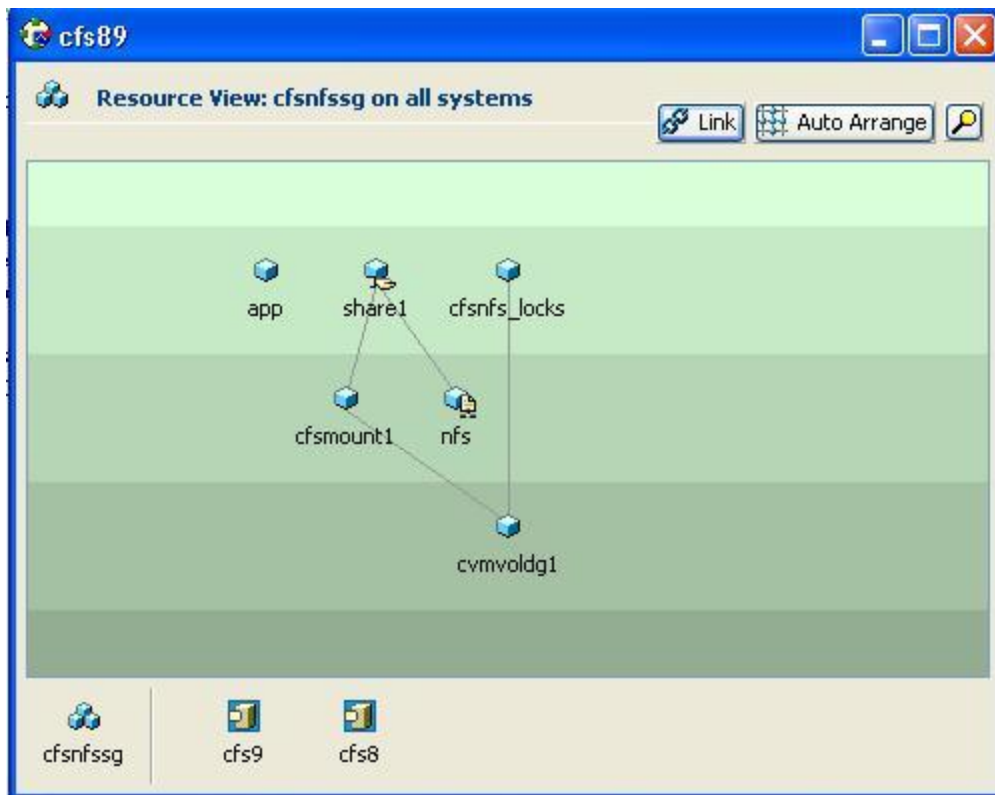


Figure 4: NFS on CFS Service group

A virtual IP service group is created to monitor the external IP for NFS clients to connect to. This service group is a failover service group and VCS is responsible to offline and online of the resources.



Figure 5: Virtual IP Service group

Summary

Symantec Cluster File System provides an efficient solution for providing active-active NFS serving at a fraction of the cost of high end NAS heads and filers. It takes advantage of existing SAN infrastructure and scalability can be achieved both at the client connectivity layer and the backend storage layer. CFS is tuned to handle multiple types of workload from access to large files to many clients accessing multiple small sized files.

Appendix

Veritas Cluster Server configuration file: main.cf

The configuration file included below can be used to rebuild or study the 2-node Clustered NFS described previously in this document.

```
include "types.cf"
include "ApplicationNone.cf"
include "CFSTypes.cf"
include "CVMTypes.cf"

cluster cfs89 (
    UserNames = { admin = cJJdJHiQJlJR }
    Administrators = { admin }
    HacliUserLevel = COMMANDROOT
)

system cfs8 (
)

system cfs9 (
)

group cfsnfssg (
    SystemList = { cfs8 = 0, cfs9 = 1 }
    AutoFailOver = 0
    Parallel = 1
    AutoStartList = { cfs8, cfs9 }
)

ApplicationNone app (
    MonitorProgram = "/opt/VRTSvcs/bin/ApplicationNone/lockdstatdmon"
)

CFSMount cfsmount1 (
    Critical = 0
    MountPoint = "/test1"
    BlockDevice = "/dev/vx/dsk/dg1/vol1"
    NodeList = { cfs8, cfs9 }
)

CFSMount cfsnfs_locks (
    Critical = 0
    MountPoint = "/locks"
    BlockDevice = "/dev/vx/dsk/dg1/vollocks"
    NodeList = { cfs8, cfs9 }
)

CVMVolDg cvmvoldg1 (
    Critical = 0
    CVMDiskGroup = dg1
    CVMActivation @cfs8 = sw
    CVMActivation @cfs9 = sw
    CVMStartVolumes = 1
)

NFS nfs (
)

Share share1 (
    PathName = "/test1"
)
```

```

requires group cvm online local firm
cfsmount1 requires cvmvoldg1
cfsnfs_locks requires cvmvoldg1
share1 requires cfsmount1
share1 requires nfs

// resource dependency tree
//
//      group cfsnfssg
//      {
//      ApplicationNone app
//      CFSSMount cfsnfs_locks
//      {
//      CVMVolDg cvmvoldg1
//      }
//      Share share1
//      {
//      NFS nfs
//      CFSSMount cfsmount1
//      {
//      CVMVolDg cvmvoldg1
//      }
//      }
//      }
//      }

group cfsnfssg_dummy (
    SystemList = { cfs8 = 0, cfs9 = 1 }
    AutoFailOver = 0
    Parallel = 1
    AutoStartList = { cfs8, cfs9 }
)

requires group cvm online local firm

// resource dependency tree
//
//      group cfsnfssg_dummy
//      {
//      }
//      }

group cvm (
    SystemList = { cfs8 = 0, cfs9 = 1 }
    AutoFailOver = 0
    Parallel = 1
    AutoStartList = { cfs8, cfs9 }
)
CFSfsckd vxfsckd (
    ActivationMode @cfs8 = { dg1 = sw }
    ActivationMode @cfs9 = { dg1 = sw }
)

CVMCluster cvm_clus (
    CVMClustName = cfs89
    CVMNodeId = { cfs8 = 0, cfs9 = 1 }
    CVMTransport = gab
    CVMTimeout = 100
)

CVMVxconfigd cvm_vxconfigd (
    Critical = 0
    CVMVxconfigdArgs = { syslog }
)

```

```

cvm_clus requires cvm_vxconfigd
vxfsckd requires cvm_clus

// resource dependency tree
//
//     group cvm
//     {
//         CFSfsckd vxfsckd
//         {
//             CVMCluster cvm_clus
//             {
//                 CVMVxconfigd cvm_vxconfigd
//             }
//         }
//     }

group vip1 (
    SystemList = { cfs8 = 0, cfs9 = 1 }
    AutoStartList = { cfs8, cfs9 }
    PreOnline @cfs8 = 1
    PreOnline @cfs9 = 1
)

IP vip1 (
    Device = bge0
    Address = "10.182.111.161"
    NetMask = "255.255.240.0"
)

NIC nic1 (
    Device = bge0
)

requires group cfsnfssg online local firm
vip1 requires nic1

// resource dependency tree
//
//     group vip1
//     {
//         IP vip1
//         {
//             NIC nic1
//         }
//     }

```


About Symantec

Symantec is a global leader in providing security, storage and systems management solutions to help businesses and consumers secure and manage their information. Headquartered in Cupertino, Calif., Symantec has operations in 40 countries. More information is available at www.symantec.com.

For specific country offices and contact numbers, please visit our Web site. For product information in the U.S., call toll-free 1 (800) 745 6054.

Symantec Corporation
World Headquarters
20330 Stevens Creek Boulevard
Cupertino, CA 95014 USA
+1 (408) 517 8000
1 (800) 721 3934
www.symantec.com

Copyright © 200 Symantec Corporation. All rights reserved. Symantec and the Symantec logo are trademarks or registered trademarks of Symantec Corporation or its affiliates in the U.S. and other countries. Other names may be trademarks of their respective owners.

05/09